

Assessment of an Open-Source Aircraft Noise Prediction Model Using Approach Phase Measurements

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An open-source simulation model for aircraft noise prediction is presented and validated using backpropagated noise measurements for a state-of-the-art engine and aircraft. The validation is focused on approach procedures and was performed using ground-based noise measurements that were taken at 17 recording stations for a total of 18 consecutive flights carried out during the morning of 8 April 2021. The flights were performed using two A321neo aircraft with LEAP-1A engines. It is demonstrated that the presented noise model provides a satisfactory estimation of the source noise for varying approach configurations and flight conditions. Configurations using a greater number of high-lift devices are particularly well predicted in the mid- and high-frequency regions, whereas the lower configuration settings show greater spectral deviations, which are partly attributed to measurement uncertainties caused by the increased aircraft–microphone distance. The model can predict the overall mean total sound intensity level within a 2 dB accuracy for all configurations, while the average predicted level at each microphone differs by less than 3 dB from the measurement average, for all cases except one. Variation in aircraft speed showed to have a strong impact on the predicted total noise, which matches the well-recognized sixth-power Mach number far-field sound intensity scaling law for airframe noise models, while the measurements indicated a less significant dependency. This is mainly due to installation effects and noise reduction measures that are not included in the models. Nevertheless, the variations in the spectra of the predicted and measured noise showed similar patterns.

Nomenclature

c_0	=	speed of sound, m/s
D	=	drag, N
g	=	gravitational acceleration, m/s ²
L	=	lift, N
M	=	Mach number
m	=	aircraft mass, kg
T	=	thrust, N
T_S	=	static atmospheric temperature, K
T_0	=	reference atmospheric temperature, K
V	=	aircraft speed, m/s
V_y	=	vertical component of aircraft speed, m/s
α	=	angle of attack, deg
γ	=	flight-path angle, deg
θ	=	pitch angle, deg
φ	=	bank angle, deg
ψ	=	aircraft heading, deg

1. Introduction

AIRCRAFT noise pollution is a major concern, especially in areas in proximity to airports. In recent years, the effects of noise on human health have become an area of intensive research. The main detrimental effect of aircraft noise is on human health. For instance, its relationship with heart disease and stroke has been well established [1]. Other known effects are hypertension [2]; effects on children's learning [3]; and, more generally, sleep disturbance and community annoyance. Growing knowledge in this area has led to the development of new guidelines, such as the recently published World Health Organization document [4]. Strong recommendations are given to reduce aircraft noise to day–night yearly averaged sound pressure level (L_{den}) below 45 dB. This should be compared with the current Swedish guidelines [5], which state 55 dB.

Efforts to reduce aircraft noise have mostly been driven by airport noise limits and regulations based on the International Civil Aviation Organization (ICAO) guidelines [6]. As a result of the former, these efforts are mainly concerned with designing or optimizing procedures, which can be directly applied using the existing fleet. The latter draws the focus on reducing the noise at the source by developing new engine and aircraft designs. In both of these cases, understanding the contribution from different airframe and engine components to the total noise level will help to reduce the noise by taking measures to improve the sources that have a larger effect.

Several aircraft noise prediction models and tools exist and are widely used. These are usually separated according to their modeling approach, as discussed in several studies [7–10]. Some of the most common models are the European Civil Aviation Conference (ECAC) Doc 29 [11] and Aviation Environmental Design Tool (AEDT) of the Federal Aviation Authority [12], which predicts the noise level according to ECAC Doc 29 or ICAO Doc 9911 [13]. These tools are based on what is commonly referred to as integrated methods, which use databases and consider the aircraft as a whole. They provide a good prediction for long-term averages, but they do not provide any information about the contribution of individual noise components.

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Another type of aircraft noise prediction that can help understand the noise source breakdown is based on scientific models. These can be separated into physics-based methods and empirical and semi-empirical correlations. The former category concerns high-fidelity methods that are based on physical laws. These methods can be applied to any design problem, and are able to provide accurate estimations, and assess the effects of uncertainties [14]. The drawback, however, is that they usually require complex and computationally expensive calculations. This approach can be partly found, for example, in SIMUL [9], which is also based on empirical methods. Finally, correlations that belong to the semi-empirical modeling category are, for example, implemented in ANOPP [15] and PANAM [16], which are well documented in the public literature but are generally not available to the public. In these tools, each source is modeled separately based on correlations developed from databases and approximations of physical models of sound generation. They require more detailed input data than the integrated methods but provide a more accurate representation of the source. Their accuracy, however, is highly dependent on the database they were developed from. Hence, their application in conceptual design may have some limitations. For this reason, these models have been continuously updated based on new databases and even enhanced with capabilities to quantify their uncertainties and are, therefore, commonly implemented in low-noise aircraft design problems [17,18]. It becomes clear that each type of aircraft prediction model has its own advantages and limitations, and selection of the appropriate tool depends on the intended application, as has been shown by the various studies in the public literature [19–24].

This work describes a new implementation, the Chalmers Noise Code (CHOICE), which is based on empirical and semi-empirical models available in the public literature. Many of the methods are common to the models integrated in ANOPP or ANOPP2. CHOICE is an open-source framework [25] with the capability to predict the source noise level, for every frequency and longitudinal directivity, from individual airframe and engine components and the entire aircraft. In this study, CHOICE was validated by comparing the noise source estimation from specific trajectories with corresponding measurements on the ground. Validation is performed for approach trajectories using flight data recorder (FDR) data and noise measurements, and the code, together with the aircraft modeling data, is available for download [25].

A number of studies on validation procedures can be found in the literature. These range from comparisons with flyover measurements to benchmark testing of auralization tools. One study was conducted by Giladi and Menachi [26]. Their study focused on validating the AEDT noise model using specific flight paths and the corresponding noise measurements. They compared the measured and predicted sound exposure level (SEL) and the maximum sound level (L_{Amax}) for several aircraft types at three airports. Jäger et al. [27] developed a noise simulation tool that stands between the two prediction approaches mentioned previously. A brief description of the model, which is based on linear regression equations, is provided along with a comparison, for several flights, of the measured and estimated A -weighted maximum sound level ($L_{AS,max}$) and the A -weighted SEL integrated over the time interval during which the sound level was above a set threshold ($L_{AE,tg}$, where tg is the time interval of integration). Filippone et al. [28] presented a validation case for a parameterized noise prediction model using departure flyover measurements for several aircraft types. The comparison of the simulated noise with the measurements on the ground was performed using the A -weighted sound pressure level (L_A or L_{AS}) or the equivalent sound level (L_{Aeq}). Additional studies have been conducted following similar validation procedures [16,29–31]. All of these studies indicated good agreement between the measured and predicted noise levels. However, the focus of the validation is on the predicted noise on the ground, which does not show any details regarding the frequency characteristics of the noise and provides no information to the reader about the source or subcomponent contribution. Other recent efforts have focused on the assessment of auralization tools through benchmark testing by defining different test cases for departure and approach, and comparing the results from various already validated tools [32].

A different approach was followed by Vieira et al. [33]. The semi-empirical model source prediction was compared with backpropagated flyover measurement data for three aircraft for takeoff and landing conditions. The noise measurements were performed over 3 days using a microphone array placed along the runway line. The flight conditions were determined using an Automatic Dependent Surveillance—Broadcast (ADS-B) receiver or through consecutive frames from an optical camera at the center of the array, while the fan rotational speed was estimated from the spectrum obtained from each flyover measurement. The specific measurement arrangement created some limitations on the variation of the flyover operating conditions, especially for landing, as the aircraft followed the instrument landing system (ILS), resulting in similar procedures. As a consequence, there is a low variation in the measured noise, and the average values of the altitude, speed, and fan rotational speed were used for the prediction and comparison. Despite the low variability in the noise spectra and evaluated flight conditions, the comparison, together with a sensitivity analysis of the different models, indicated some frequency regions where deviations between the prediction and experimental data occurred. These deviations could be partly explained by the fact that their model neglects the contribution from some important components, such as the turbine, which, for current engines, can have a significant impact on the total noise [34,35]. Furthermore, various studies have shown that, during approach, the airframe and fan are the dominant sources of noise [34,36,37], followed by the turbine, and finally the jet and combustor. However, interestingly, in their model, combustor noise seems to be one of the driving components.

A similar comparison approach was followed for a newer aircraft and a more complete simulation model. The establishment of the simulated engine can be found in [35], with a comprehensive comparison against official public data of the LEAP-1A engine in terms of performance, NO_x , and noise emissions. Among them, noise predictions using the presented tool, compared to the certification results found in the European Union Aviation Safety Agency (EASA) noise database, are considerably well matched. The noise source breakdown is also presented for the various certification points, indicating that engine noise is dominant during takeoff, but its contribution decreases during approach. In the present study, validation was performed using FDR approach trajectories, which increased the accuracy of the flight parameters that were input to the models. Eighteen approach flights were performed using two A321neo with a LEAP-1A engine at Arlanda Airport in Stockholm, Sweden. The purpose of these flight tests was to collect noise measurement data for several approach procedures that differ in terms of speed, approach altitude, and landing gear deployment position. These planned variations followed the standards of the normal ILS approach. The sound intensity level (L_I) was computed at the source for every frequency and longitudinal directivity along the trajectory and compared with the backpropagated noise measurements from several microphones underneath the aircraft flight path. This method of validation allows for understanding the individual contributions of the sources to the total aircraft noise for varying configurations and flight conditions that are found during normal approach procedures. The results for each configuration are presented separately. This enables the assessment of not only the impact of each configuration change (extension of the slats and flaps or landing gear deployment) on the noise but also of the effect that variation in flight parameters has on the noise within the same configuration. For example, the effect of speed for configurations with a low degree of high-lift device deployment might not be as prominent as that for the landing configuration. This kind of assessment is important to consider in the design of new procedures.

The analysis starts with an introduction to the noise prediction model and the implemented methods in Sec. II, followed by a description of the available data and the noise source estimation in Secs. III and IV, respectively. The validation process is presented in detail in Sec. V, and the results are demonstrated for every high-lift device and landing gear setting, and the dependence of the source noise level on the aircraft speed is also examined. A sensitivity analysis of the noise prediction model for variations in flight parameters is presented in Sec. VI. Finally, in the Appendix, the detailed spectra of the source noise level

are provided, where the individual component contributions for every configuration are visible.

II. Noise Model

The noise estimation was based on empirical and semi-empirical noise source models found in the public literature. Every engine and airframe component was modeled separately, and the total aircraft noise was calculated as the sum of all the individual components. The noise source models were briefly described in [35], but the noise prediction was updated to account for the effect of the suppression mechanisms. A summary of the implemented models, with more details on the required input parameters, is presented in this section.

A. Fan and Compressor Noise

The model implemented for the prediction of noise from the fan and compressor was based on the method introduced by Heidmann [38] and later updated by Kontos et al. [39]. The predicted noise is in one-third octave-band frequencies of the free-field noise pattern, which, for the fan or compressor inlet duct, consists of broadband, discrete, and combination tone noise, and for the fan discharge duct consisted of broadband and discrete tone noise. The parameters required to predict the basic spectrum levels are the mass flow rate, total temperature rise for a fan or compressor stage, and design and operating point values of the rotor-tip relative inlet Mach number.

B. Core Noise

The combustor noise source model was based on the method described by Gliebe et al. [36,40] for low-emission combustors. Far-field noise data were collected and analyzed for both single annular and dual annular combustors, and two correlations were developed, accounting for the combustor geometry, cycle conditions, spectral frequency content, and directivity. The spectral noise peaks for single annular combustors were observed at three frequencies and directivities: 63, 160, and 630 Hz and 150, 130, and 130 deg, respectively, whereas for the dual annular combustors, they were found at 160 and 500 Hz and 130 deg. The peak overall sound pressure level and normalized sound pressure level are computed for the stated peak frequencies and directivities, and the final spectrum is determined for every frequency and directivity by combining the two and an air attenuation factor.

C. Turbine Noise

Turbine noise was estimated based on the model described by Dunn and Peart [41] and consists of broadband and discrete tone noise components, which are related to the last stage relative tip velocity, primary mass flow, and local speed of sound at the turbine exit. The discrete tone noise is also dependent on the stator/rotor spacing. The spectra for both components are normalized with respect to the fundamental blade passage frequency of the last stage of the turbine and are given in one-third octave-band levels at the free-field index ($R = 1$ m) conditions.

D. Jet Noise

The jet noise model was presented by Russell [42]. This method was based on extensive test data for coaxial and circular jets. The data were curve fitted as bicubic splines using the least-squares method as a function of frequency and directivity. A third-order Taylor series was used for the curve fitting. The coaxial jet noise is calculated as a function of the jet state properties for a single-stream equivalent jet with the same mass flow, energy flow, and thrust as the coaxial jet. The one-third octave-band sound pressure level is calculated as the summation of four components: normalized overall power level, directivity index, power spectrum level, and relative spectrum level.

E. Airframe Noise

The airframe noise was modeled as the sum of the independent nonpropulsive noise sources of the aircraft. These are the clean wing and tailplanes, high-lift systems (flaps and slats), and landing gear.

The method proposed by Fink [43,44] was implemented for wings, tail surfaces, and high-lift components. According to this, the clean wing and horizontal and vertical tails radiate noise as a result of the turbulent boundary layers at the trailing edges. The overall sound pressure level is modeled as a function of aircraft speed, boundary-layer thickness, and wing or tail span. The prediction refers to aerodynamically clean sailplanes, and an additional 6 dB term was included for the wing, as suggested for jet aircraft. The spectrum shape is then calculated using a semi-empirical equation for the normalized spectral density of the trailing-edge noise from externally blown flaps. The deflection of flaps causes a noise increment, which is modeled as a function of flap area, flap deflection angle, aircraft speed, and Strouhal number calculated for the flap chord. For leading-edge high-lift devices, in this case slats, a simple approach is used, where it is assumed that the generated noise is the sum of the clean wing noise for jet aircraft increased by 3 dB and a higher-frequency trailing-edge noise of the same amplitude scaled relative to the slat chord. The total noise spectrum is then calculated by summing the components as uncorrelated sources, and a 3 dB reduction is applied to the spectrum to reach free-field conditions.

The landing gear noise was modeled according to the method proposed by Sen et al. [45]. This method assumes that the noise generated from a landing gear can be decomposed into four components: low-frequency component generated by the struts, mid-frequency component caused by the hydraulic pipes near the break assembly, high-frequency component from electric wires and small pipes, and tire noise component. The local flow Mach number was set equal to 75% of the flight Mach number instead of 83%, as suggested in the report, because it was found to better match the measurement data. This value is within the valid range [46–48]. The total landing gear noise was computed by summing all components and depends on the aircraft Mach number, emission angle, and length scale for each component.

F. Acoustic Treatment

The methods implemented for the fan and turbine noise prediction resulted in an overprediction of the noise level. When looking at the fan, this overprediction can be attributed to the fact that the implemented method [38] is a hard wall prediction, and no acoustic treatment is considered. Another reason is that the method was developed based on static fan tests. Such noise data are often affected by the noise caused by the ingestion of vortices or other turbulence from the test stand or other nearby structures, as suggested by Krejsa and Stone [49]. The implemented turbine noise model [41] was evaluated and compared with other methods by Krejsa and Valerino [50]. It was found that when using turbine noise measurement data, the prediction showed a significant deviation ranging from –6 to 8 dB. This deviation can be partly attributed to acoustic treatment, which is not included in the model. Attempts to model this by introducing a turbine work term have not been successful. Despite this deviation, the authors compared different methods found in the public literature and recommended the method by Dunn and Peart [41]; however, the method should be reevaluated and modified as more data become available. Following this comment, no further updates or more recent correlations have been found.

A typical example of the effect of acoustic treatment on fan and turbine noise is presented in Fig. 1, where the contributions of each component to the total noise level are also evident. This figure was originally introduced by Owens [51], and later used in a study by Groeneweg and Rice [37], where a thorough discussion was presented on the different mechanisms involved in fan noise generation and suppression. Further discussion on aircraft noise reduction technologies can be found in the bibliographic review presented by Casalino et al. [52] and in the overview presented by Leylekian et al. [53]; however, it is out of the scope of this study to discuss the reduction mechanisms in detail.

For the reasons described previously, a noise suppression module was implemented and applied to the fan inlet, fan discharge, and turbine noise components. This module was based on the suppression

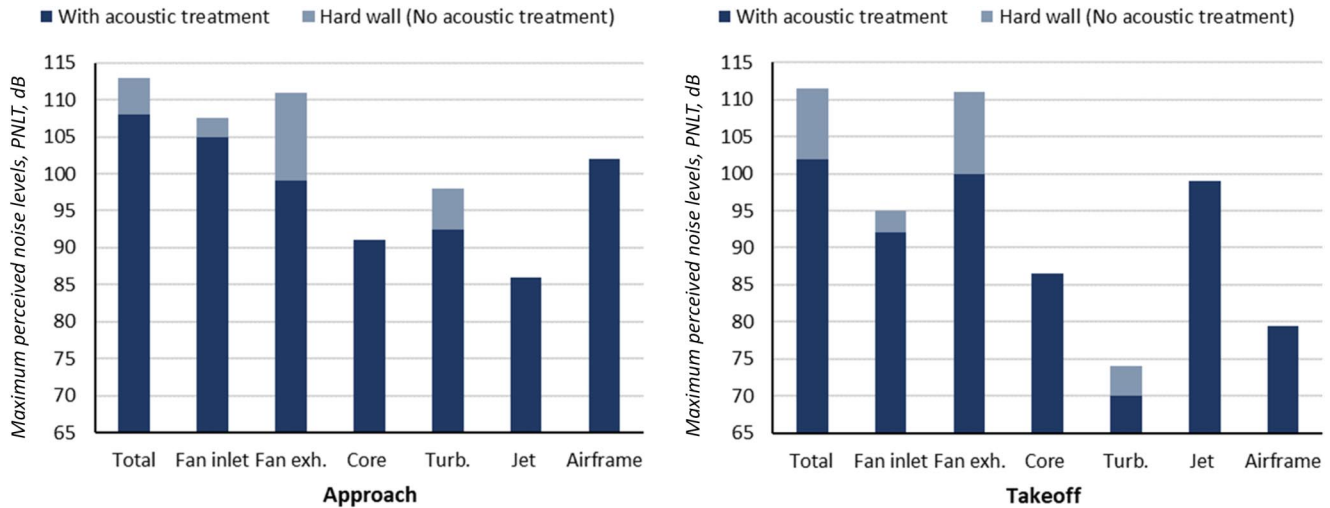


Fig. 1 Effect of acoustic treatment on engine components. (Reproduced from [51]). (Fan exh., Fan exhaust; Turb., Turbine).

module, implemented in ANOPP, where a suppression factor is assumed and applied for every frequency and directivity, as described by Wilson [54].

In this study, the same suppression factor was assumed for each frequency and directivity. The choice of values was based on the suggested values by Willshire and Garber [55] and adapted for a more recent engine. More specifically, suppression values of 10, 15, and 15 dB were applied to the fan inlet, fan discharge, and turbine noise components, respectively. These values follow the assumptions presented by Willshire and Garber [55], and an additional 5 dB suppression was added to the inlet fan, as it has been shown to be overpredicted by the implemented model [49].

III. Data Availability and Processing

The validation process was based on FDR and noise measurement data that were collected within the Approach Noise Trials project [56] during the morning of 8 April 2021. The flights were conducted on Runway 26 at Arlanda Airport purely for noise measurement. Approximately 25 microphones were used, located along a 15 n mile approach path and at a height of 1.2 m above ground. During the flights, after the aircraft passed the last microphone, 1.6 n mile before the runway threshold, it turned around to perform a new approach until all the planned flights were performed. A detailed description of the measurement setup was provided by Åbom et al. [56]. Data from 17 microphones were used for the present validation process. The positions of the microphones are shown in Fig. 2. Arlanda Airport can be seen at the bottom left of the figure, and the ground track that all

flights followed is indicated by a white line. For the rest of the analysis, when referring to a specific microphone, the corresponding number from Fig. 2 is used.

The recorded noise data were logged in one-third octave-band levels with a 125 ms update frequency. After postprocessing, the noise data were converted into A-weighted sound pressure levels and stored with slow time weighting equal to 1 s. The measurements refer to the ground, and using backpropagation, the noise source level was estimated. The backpropagation included spherical spreading, frequency-dependent sound absorption, effect of changed density and temperature with altitude, and removal of ground reflection. The latter is usually the most uncertain part of any atmospheric propagation process. The method used to remove the ground reflection effect from the data was presented by Chessell [57]. According to this method, the reflection of spherical waves at the boundary was considered, and the acoustic properties of fibrous absorbent materials was used to model the essential properties of real soil surfaces. This method is an expansion of the original work by Delany and Bazley [58,59], where they showed that the characteristic impedance of the material can be modeled as a function of frequency divided by specific flow resistance. In this study, the flow resistance was assumed equal to 200 cgs for grass-covered ground [60]. The noise measurement data processing and backpropagation were performed by Tengzelius and Johansson, and a more detailed description is reported in the project's webpage [61].

From the available parameters in the FDR data, only a down-selection were used. These included trajectory details (aircraft position and altitude), Mach number and vertical speed, static temperature, flap and slat setting, pitch and roll angles, and wind

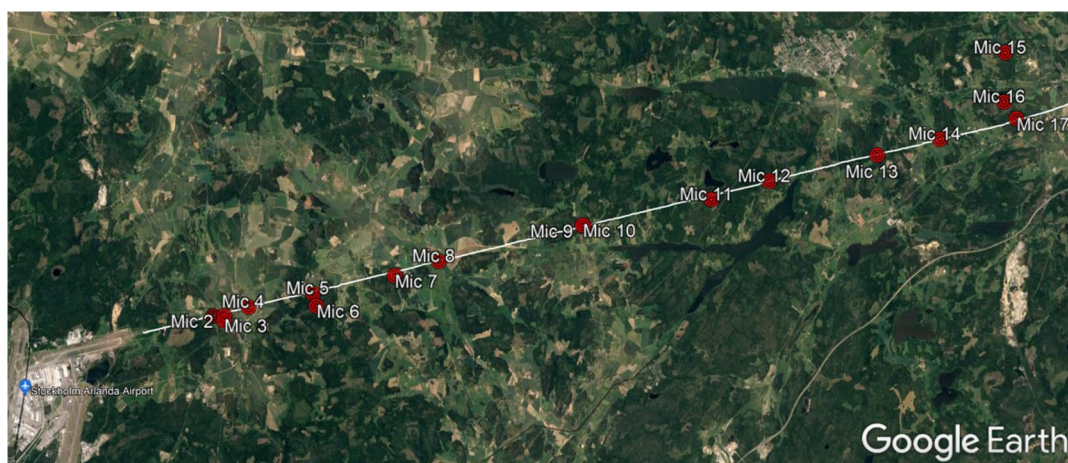


Fig. 2 Microphone numbers and positions.

speed and direction. From these parameters, it was possible to determine the true airspeed, flight-path angle, angle of attack, and acceleration of the aircraft, and predict the thrust and aerodynamic performance. More specifically, the following equations were used:

$$a = \theta - \gamma \quad (1)$$

where θ is the pitch angle, and γ is the flight-path angle calculated from the vertical speed and true airspeed, as follows:

$$\gamma = \sin^{-1}\left(\frac{V_y}{V}\right) \quad (2)$$

with $V = c_0 M \sqrt{T_s/T_0}$.

To estimate the aerodynamic performance of the aircraft, the lift-to-drag ratio, which was derived from an available aircraft model for the A321-231, was used. This value varied for each configuration and landing gear position. The thrust T , lift L , and drag D were then determined using the three-dimensional flight dynamics equations, as indicated by the following system of equations:

$$\begin{cases} T \cdot \cos(a) - D - m \cdot g \cdot \sin(\gamma) = m \cdot \dot{V} \\ (T \cdot \sin(a) + L) \cdot \cos(\varphi) - m \cdot g \cdot \cos(\gamma) = m \cdot V \cdot \dot{\gamma} \\ (T \cdot \sin(a) + L) \cdot \sin(\varphi) = m \cdot V \cdot \cos(\gamma) \cdot \dot{\psi} \end{cases} \quad (3)$$

where a is the angle of attack, m is the aircraft mass, φ is the bank angle, and ψ is the heading angle.

IV. Noise Source Estimation

The engine noise source estimation in CHOICE requires several engine performance parameters, which are estimated using the in-house code GESTPAN [62]. GESTPAN takes thrust, airspeed, altitude, and ambient conditions as input and generates several performance files for each of the engine components.

The airframe noise model required the geometric characteristics of several components. These were obtained either from scaled drawings of the aircraft [63] or from an open-access database [64]. For flap noise prediction, the noise was modeled as the contribution of four flap elements with dimensions derived from the scaled drawing [63]. For the landing gear, the wheel diameters were obtained from the aircraft characteristics report [63], whereas the typical dimensions of the main struts and small parts were approximated from the respective data provided by Guo [47,48] for the Boeing 737 and 777 landing gears. These dimensions were provided to CHOICE through a separate input file.

The performance and dimension files for the engine and airframe were then input to CHOICE together with a trajectory file, and the aircraft noise was calculated for every point in the trajectory. The output from CHOICE was a sound pressure level matrix for every frequency and longitudinal directivity and for every trajectory point. This matrix was generated for every component separately, and the total noise was calculated by summing all the individual sources.

V. Validation

For the validation, the noise source estimates were divided according to the aircraft configuration. The configuration settings for the A321neo are listed in Table 1. It should be noted that

Table 1 A321neo configurations

Configuration	Slats, deg	Flaps, deg	Landing gear
Clean	0	0	Up
1	18	0	Up
1+F	18	10	Up
2	22	14	Up/down
3	22	21	Down
FULL	27	35	Down

configuration 1+F does not appear in the measurements because it is mainly used for takeoff. Configuration 3 and FULL are always with landing gear extended, while configuration 2 can be with landing gear retracted or extended.

Validation was performed by comparing the source noise spectra directly underneath the aircraft at $\theta = 90$ deg and $\varphi = 90$ deg, where θ and φ are the longitudinal and lateral directivity. Because the aircraft was performing a normal ILS approach, a descent angle of 3 deg was used. Thus, the longitudinal directivity of 90 deg does not correspond exactly at the overhead point but rather a bit before. Because of the nature of the measurements, the noise level at that point is not always measured; thus, interpolation was performed to obtain the spectrum at the desired point. First, the source noise level was compared for every frequency. For each configuration, the mean spectra at each microphone and the overall mean are presented. This means that if the aircraft passed above a microphone with the same configuration several times, then the average of the spectra from all passages was calculated. This way, the statistical variation of the noise measurements was minimized. This variation is one of the most common problems in noise measurements, and it can occur even for the same aircraft type, flight procedure, and environmental conditions [26,65]. This can also be observed from the figures in the report by Tengzelius and Johansson [61], where the total sound intensity level at the source was plotted for various speed, thrust, and rotational speed settings. It is shown that for the same setting, the intensity level can vary by as much as 10 dB. Finally, for the validation, only the microphones that were directly overflown were considered, that is, measurements with lateral directivity smaller than 80 deg were omitted.

The spectra for all configurations are shown in Figs. 3–8, following the order in which the configurations are presented in Table 1. The measurement data are depicted with variations of the red color and the model estimations with blue color variations. Each microphone is represented by a dashed, dotted, or dashed-dotted line that differs in color variation and thickness from the others. The overall mean is indicated by the thick solid line.

In Fig. 3, the spectra for the Clean configuration are presented. The noise level is generally well predicted, especially in the mid-frequency region, with both the spectrum shape and the level of prediction matching the measurements. Both spectra suggest an increase in the high-frequency region mainly driven by the fan. The prediction also shows a peak from the fan at 1 kHz, which appears as the minimum point in the data. This deviation could be attributed to

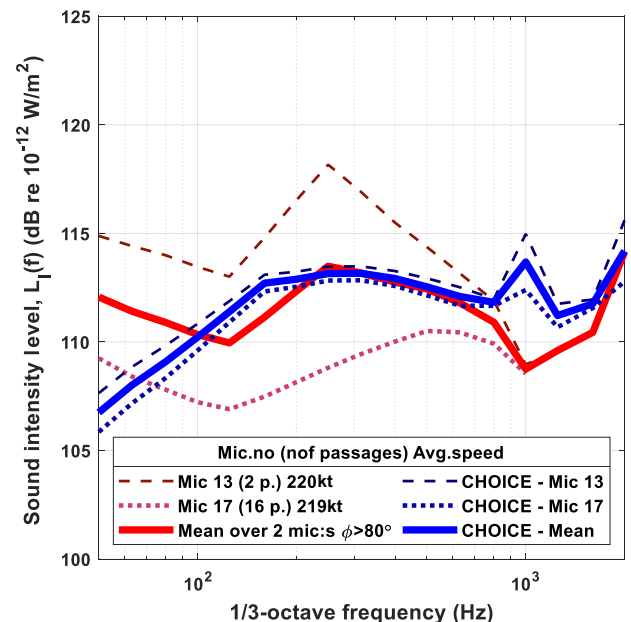


Fig. 3 Clean configuration measured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

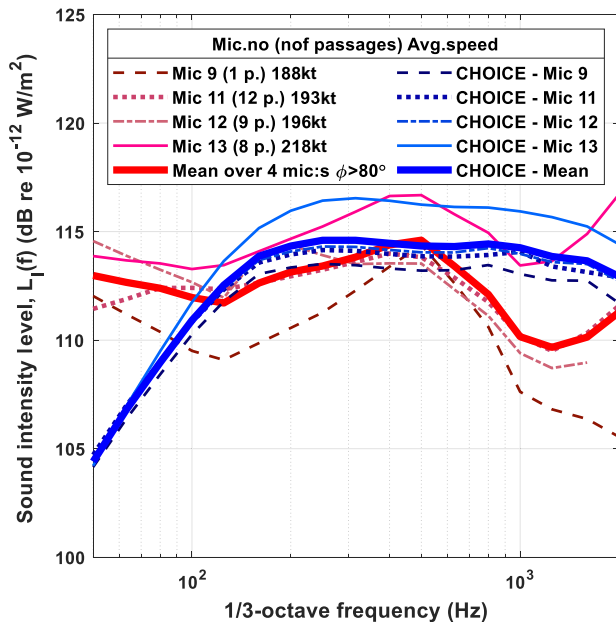


Fig. 4 Configuration 1 m easured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

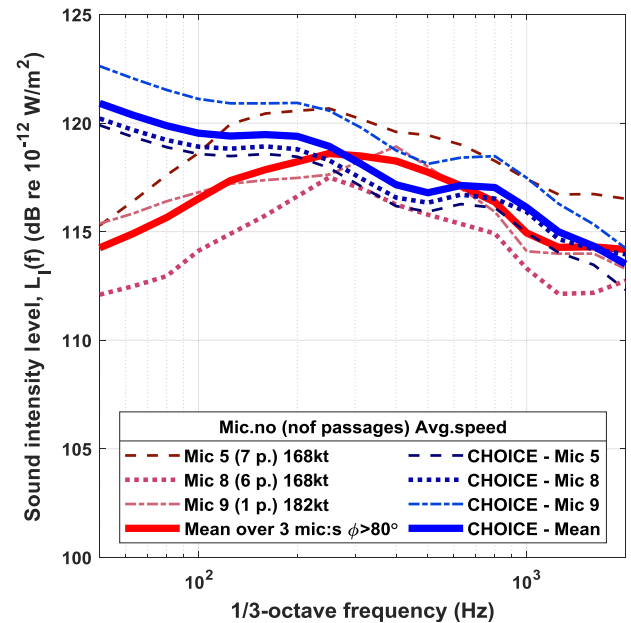


Fig. 6 Configuration 2 with the landing gear extended measured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

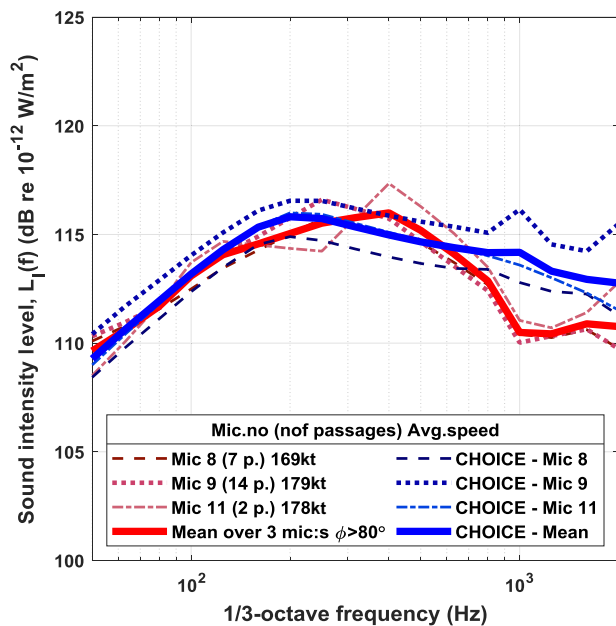


Fig. 5 Configuration 2 measured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

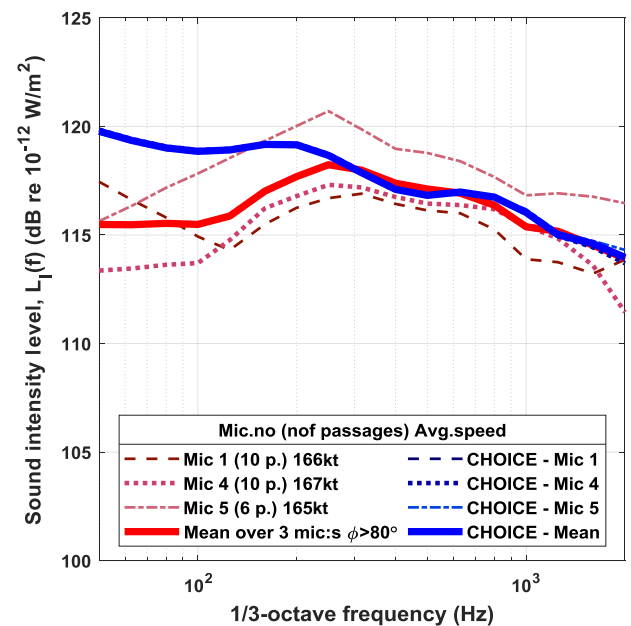


Fig. 7 Configuration 3 (always with the landing gear extended) measured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

the simplified noise suppression approximation that was implemented, which seems to lead to an underestimation of fan noise suppression for this frequency. In the low-frequency region, the two spectra deviate, with the measurement data suggesting an increase. In this region, the measurements can be significantly affected by background noise [47,66,67], which, in this case, can be caused, for example, by increased vegetation near the microphones. The sound spectra for configuration 1 are shown in Fig. 4. In this case, the agreement is not as good as for the Clean configuration, with more notable deviations appearing in the low- and high-frequency regions. The former can be attributed to increased background noise, as mentioned previously. The high-frequency region is usually dominated by engine noise, mainly from the fan, but in this case, it is masked by the airframe

noise. If closely observed, two peaks from the fan can be seen at 800 Hz and at about 1500 Hz for Mic 9. This behavior is believed to be caused by the fact that the slat noise is based on a rather simplified model, which may not capture the spectrum shape very well. In the mid-frequency region, the sound intensity level was well predicted, but there were some differences observed in the shape of the spectrum. Another plausible explanation for the discussed deviations in both the Clean configuration and configuration 1 is that during these configurations, the aircraft was farther away from the microphones, which could cause higher uncertainty in the ground measurements, increased background noise, and possible errors introduced during the backpropagation process. For configuration 2, with the landing gear retracted, as shown in Fig. 5, both the shapes and level of the spectra are

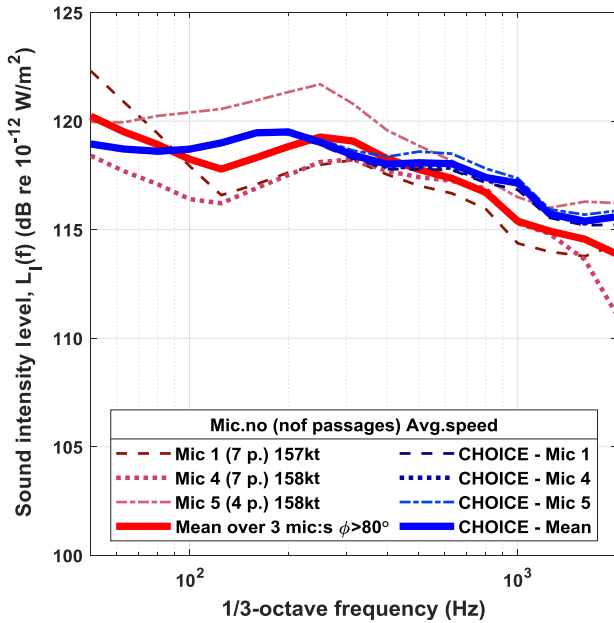


Fig. 8 Configuration FULL (always with the landing gear extended) measured and estimated (CHOICE) sound intensity level at the source (1 m); mean spectrum for each microphone and overall mean for $\theta = 90^\circ$ deg. (p., passes; nof, number of; Avg., Average; Mic., Microphone).

in good agreement, especially in the low- and mid-frequency regions. For higher frequencies, the noise was again slightly overpredicted and driven by the airframe component, with some tones appearing from the fan. In this case, the measurement data also seem to indicate airframe noise dominance, as the high-frequency noise level is relatively low and there are no peaks.

For the rest of the configurations, the landing gear was extended. The spectra for configuration 2 with the landing gear extended are presented in Fig. 6, where an evident increase is observed for the predicted spectra at low frequencies caused by the landing gear extension. However, this is not seen in the measured data. As mentioned before, this is a region of high uncertainty for both the measurements and landing gear prediction method, which was developed based on experimental data from an anechoic chamber with a cutoff frequency of 200 Hz [47]. Above this frequency, the prediction spectra showed two humps at approximately 250 and 700 Hz. These were generated from the flaps and mid-frequency component of the

landing gear, respectively. The mean measurement spectrum appears smooth, and the humps are evened out, although they can be seen in the spectra related to Mic 5 and Mic 8. Figure 7 shows the spectra for configuration 3 (always with the landing gear extended). The overall mean level was well predicted, and the shapes of the spectra suggest very good agreement for frequencies above 200 Hz. In this case, the humps can be seen more clearly in the measurement data and at frequencies very close to the ones observed from the prediction. Furthermore, in the low-frequency region, the data show a slight increase mainly driven by the spectra from Mic 1. This suggests that the location of the microphone, and hence its surroundings and distance to the aircraft, has a significant effect on this region. It is also interesting to note that for this configuration, the average speed of the aircraft when overpassing each microphone was almost unchanged, and, as expected, the spectra from CHOICE coincided with each other. On the contrary, some variation appears in the measurements, which could be attributed to the previously mentioned statistical variation. Finally, a comparison of configuration FULL (always with the landing gear extended) is presented in Fig. 8. This is believed to be the most well-predicted case. Both the shape and level of the spectra are in good agreement, and an increase in the lower frequencies was observed in both the prediction and the data. The high-frequency noise level from the prediction was slightly higher than the measurement data average, probably due to increased engine noise. However, Mic 5 also captured a higher noise level, and hence, the prediction was within the variation range of the spectra presented for the microphones.

A summary of the measured and predicted total source strengths is provided in Table 2. The total sound intensity level was calculated for every configuration by summing the level at each frequency band, for the mean spectrum above each microphone, and for the overall mean. These spectra are presented in Figs. 2–8. Overall, it can be observed that the absolute differences remain below 2 dB, for the majority of the cases with the exception of Mic 17 for the Clean configuration, Mic 8 and Mic 9 for configuration 2 with landing gear extended, and Mic 1 and 4 for configuration 3 with landing gear extended. The noise level at Mic 17 was overpredicted by the model compared to the backpropagated data. This can be seen both in Fig. 3 and Table 2. The measured noise levels at the two microphones suggest a strong variability, despite the almost constant speed. This can be attributed to the statistical variation of the data and the fact that these microphones are located relatively far from the runway threshold (Fig. 2), resulting in an increased distance between the aircraft and microphone and, possibly, in reduced accuracy, as mentioned previously. For configurations 2 and 3, both with the landing gear extended, the

Table 2 Total sound intensity level (dB re 10^{-12} W/m²) at the source, directly under the aircraft

Configuration	Microphone	1	4	5	8	9	11	12	13	17	Mean
Clean	Measured								126.7	121.0	124.0
	Model	—	—	—	—	—	—	—	124.9	123.7	124.3
	Difference								1.80	−2.78	−0.27
C1	Measured					123.1	124.6	125.0	127.2		124.9
	Model	—	—	—	—	124.5	125.2	125.3	127.2	—	125.5
	Difference					−1.39	−0.54	−0.30	0.04		−0.64
C2	Measured				125.5	125.9	126.2				125.8
	Model	—	—	—	125.4	127.4	126.2	—	—	—	126.3
	Difference				0.09	−1.50	−0.06				−0.52
C2 + LG down	Measured			131.0	127.1	128.9					129.0
	Model	—	—	129.6	130.0	132.1	—	—	—	—	130.6
	Difference			1.41	−2.90	−3.21					−1.59
C3 + LG down	Measured	127.9	127.7	130.6							128.7
	Model	130.1	130.2	130.2	—	—	—	—	—	—	130.1
	Difference	−2.14	−2.45	0.35							−1.48
FULL +LG down	Measured	130.1	129.1	131.8							130.2
	Model	130.3	130.4	130.7	—	—	—	—	—	—	130.5
	Difference	−0.20	−1.30	1.14							−0.22

The mean for each overflown microphone and the overall mean are presented.

deviation from the data was expected due to the increase in the predicted noise level in the low-frequency region, which is considered a region of high uncertainty, and no absolute conclusions can be drawn for the reasons behind the deviation. Finally, regarding the mean total sound intensity level, it is well predicted for all the configurations, whereas as expected from the analysis on the detailed spectra, the prediction for configuration FULL with the landing gear extended is particularly accurate.

To study the effect of the aircraft speed on the source noise level, three spectra are presented for each configuration, for the minimum,

maximum, and average speeds of the aircraft during the specified setting. Figures 9 and 10 depict the spectra for the configurations without landing gear (i.e., Clean configuration, configuration 1, and configuration 2), as estimated from the model and measurements, respectively. Figures 11 and 12 refer to configurations 2, 3, and FULL, where the landing gear is extended. It should be noted here that, for this part, the spectra referring to the measurements were not derived from the raw measurement data but rather from a multiple linear regression model that was developed by Tengzelius and Johansson [61] based on the data to avoid the previously mentioned

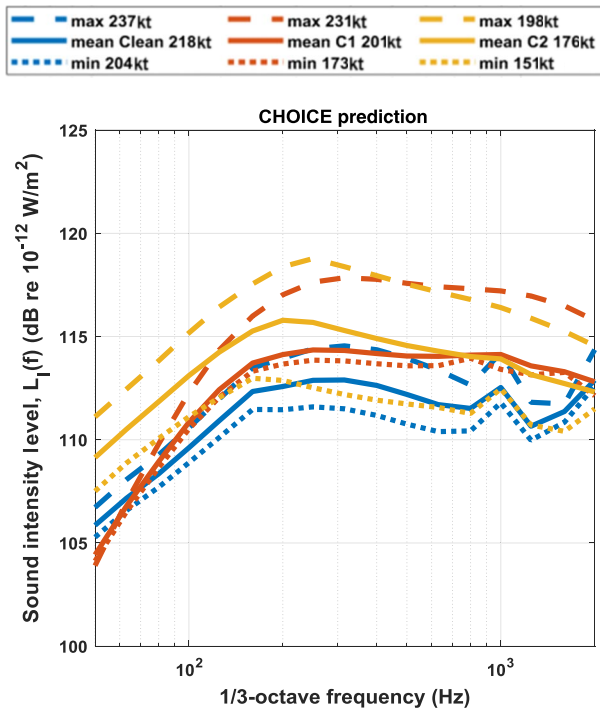


Fig. 9 CHOICE source spectra for Clean, C1, and C2 straight underneath the aircraft.

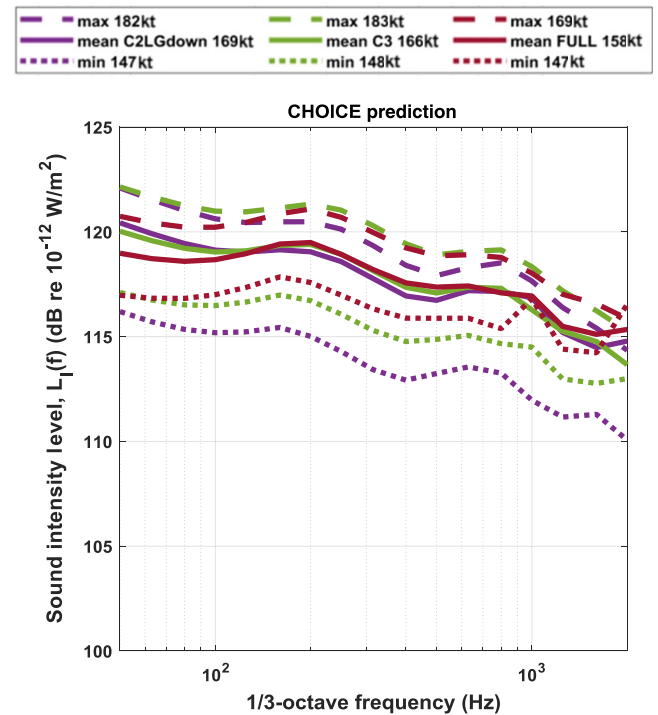


Fig. 11 CHOICE source spectra for C2, C3, and FULL (all with the landing gear extended) straight underneath the aircraft.

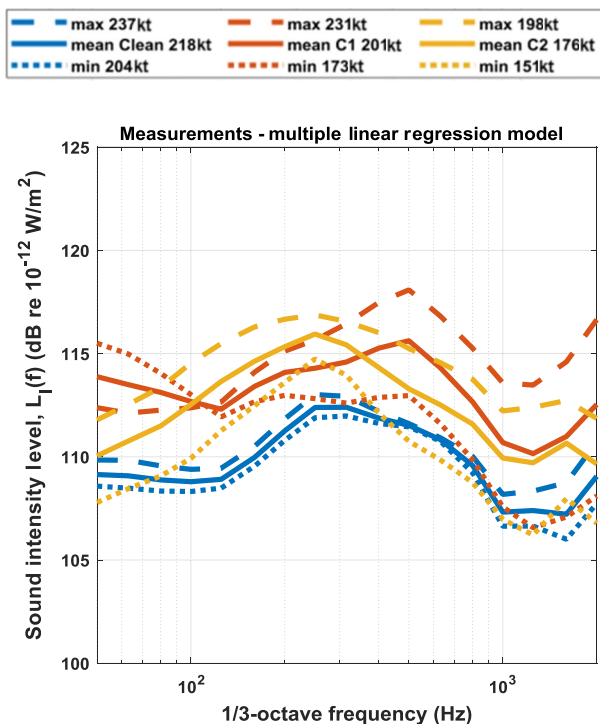


Fig. 10 Linear regression model source spectra for Clean, C1, and C2 straight underneath the aircraft.

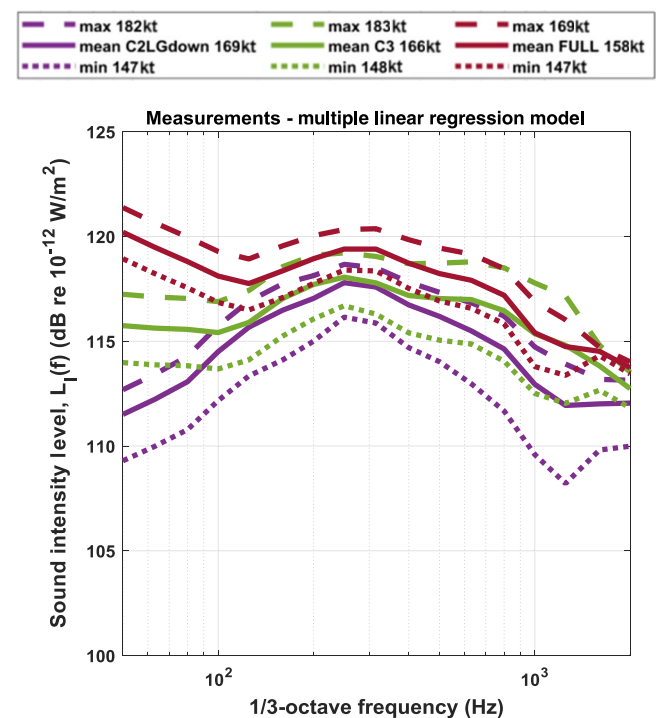


Fig. 12 Linear regression model source spectra for C2, C3, and FULL (all with the landing gear extended) straight underneath the aircraft.

statistical variation. The measurement probability distribution and multiple linear regression prediction as a function of true airspeed are presented in their report ([61] p. 22).

It can be observed that the results from CHOICE show a strong dependency on airspeed for all configurations. From the regression model, the Clean configuration and configuration 1 do not show a clear pattern with airspeed, but they indicate a slight increase in noise with increasing speed. As noted previously, these configurations correspond to aircraft positions farther away from the microphones and errors due to, for example, the backpropagation might have been introduced. For configuration 2, with the landing gear retracted and higher configurations, the airspeed dependency is evident in both the simulation and regression models. For configuration 2, with the landing gear retracted, the noise level at the average speed was well predicted, but the effect of the speed variation was slightly more evident in the prediction. A slightly stronger dependency of the simulation model on the speed was also observed for the configurations with the landing gear extended, although the general trend observed in the figure is in good agreement with the regression model. It is worth noting that for all cases, the spectra of the two models follow similar patterns. For example, in configuration 2 with the landing gear retracted, in both models, the three curves converge or overlap at low frequencies, whereas they start to deviate as the frequency increases. Similarly, configuration 2 with the landing gear extended and configuration 3 showed a constant level of difference between the different speed cases for the most part of the frequency spectrum, with a decrease when higher frequencies were reached.

So far, the focus of the study has been on the source noise prediction. The intention was to show that the effects caused by variations in configuration and approach flight conditions on the source noise level were well captured by the model. However, because the noise on the ground is usually of interest, some samples of the ground noise level vs time history accounting for the propagation effects are shown in Fig. 13, given results for selected microphone locations. For the presented cases, one flyover was selected corresponding to the approach that was expected to yield the highest noise level on the ground. More specifically, during the selected flyover, the aircraft began the approach segment at a lower altitude (2500 ft) and higher speed (220 kt). The landing gear was extended

earlier, at 6.2 n miles from the landing threshold, and the aircraft landed with configuration FULL. The microphones were selected based on Table 2, but for brevity, the results for every other microphone are shown, starting from the microphone located the farthest from the landing threshold (Mic 17) until the microphone closest to the runway (Mic 1). The results were plotted for 30 s before and after the overhead point, and a lower limit of 40 dB was set to eliminate any background noise, which was estimated to vary between 30 and 35 dB on average. An exception was observed for Mic 5, which showed a higher background noise, in the range of 35–45 dB.

The model indicated an overall satisfactory agreement with the measured noise on the ground. An overprediction was observed before the overhead point for Mic 17 and Mic 9. The noise in Mic 12 was particularly well predicted throughout the flyover, whereas in Mic 5, the noise level was underpredicted as the aircraft moved away from the microphone. This can be partly attributed to the increased background noise observed at this microphone. Mic 1 is a special case because it was located only 1.6 n mile from the landing threshold. The results are shown for a shorter time period after which the aircraft begins to climb. This microphone suggests a stronger overprediction in the forward direction, leading to a higher peak level before the overhead point, whereas the aft radiated noise was underpredicted. During this microphone flyover, the aircraft was in a constant transitioning phase as it switched from landing with FULL configuration to a sudden increase in speed and climb angle. These changes were not well captured by the models, which are usually based on static tests and time-averaged predictions. Another reason for the observed deviations could be the simplified suppression model, which assumed a constant reduction for every directivity angle and frequency. In general, the results and observations presented here are in line with an older validation study of ANOPP performed for a Lockheed L-1011 [68]. The rest of the available flyover trajectories showed a similar behavior.

VI. Sensitivity Analysis

In this section, the effect of the flight parameters on semi-empirical source noise prediction is investigated. The parameters considered for the analysis are the flight Mach number and thrust, because these

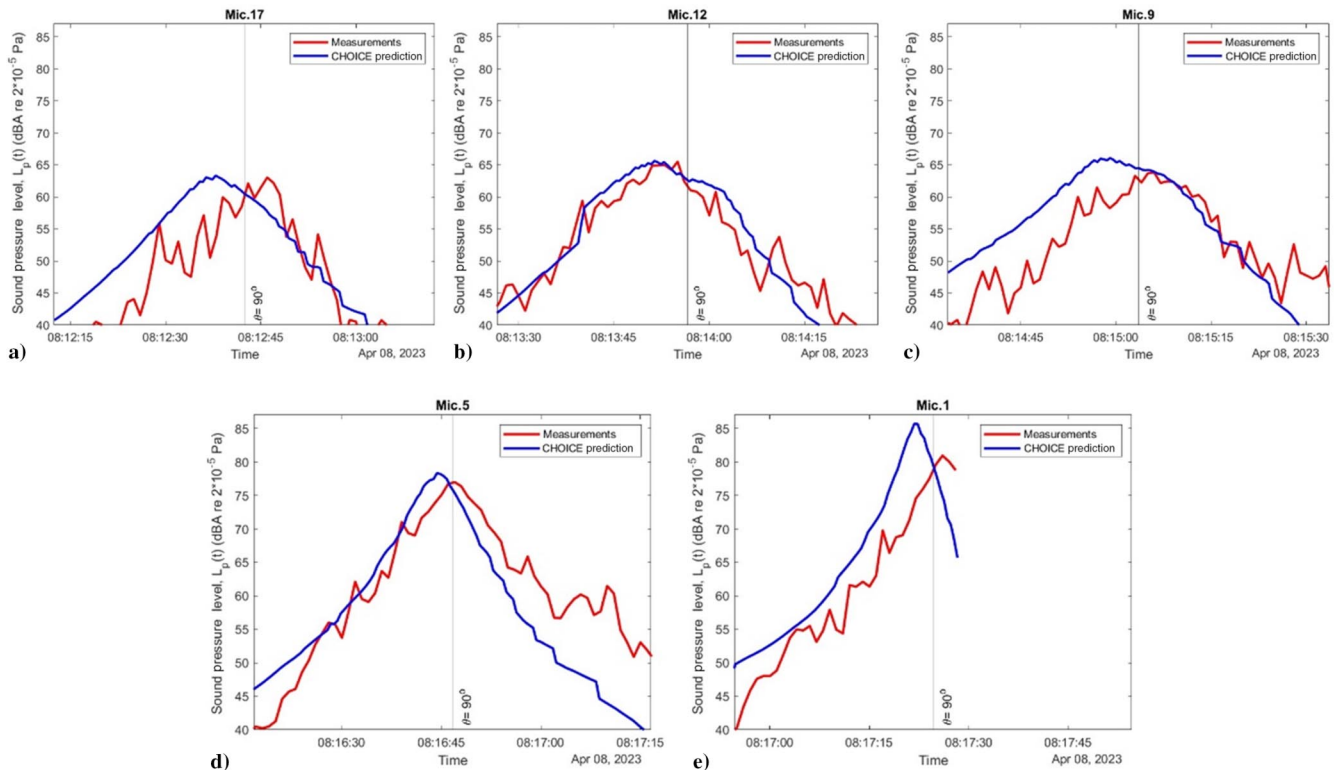


Fig. 13 Samples of the noise predictions over time history for the most critical approach procedure for a) Mic 17, b) Mic 12, c) Mic 9, d) Mic 5, and e) Mic 1.

can be directly controlled by the pilot. Variation in atmospheric conditions was not included because previous studies have shown that they do not have a significant effect on noise and can lead to a maximum deviation of 2 dB [29,61].

Figure 14 shows the variations in the Mach number and engine thrust for all aircraft configurations found in the data. The horizontal red line in each box indicates the median, and the bottom and top lines indicate the 25th and 75th percentiles, respectively. The dashed lines extend to the minimum and maximum values of the parameters that are not considered outliers, and the red crosses represent the outliers. For every configuration, the median is selected as a baseline case for each of the two parameters, and all other required parameters for noise prediction are estimated as the average of the data points whose Mach number or thrust corresponds to the median. For example, for configuration FULL with the landing gear extended, the median Mach number is equal to 0.24, and the data points that correspond to this value amount to a thrust of 67 kN and at an altitude of 287 m with an atmospheric temperature of 273 K. Note that the thrust calculated for the median Mach number does not necessarily equal the median thrust. Once the two baseline cases were defined, two studies were performed by allowing 10% variation from the median for the Mach number and thrust. This required recalculation of the engine performance for each study, while all engine design parameters were kept constant.

The results for the total aircraft noise level are summarized in Fig. 15. In the figure, the median values for the Mach number, symbolized with MM, and thrust, TM, are also presented, together with the average thrust and Mach, respectively, which were used for all three cases in the same configuration. All configurations showed the same variation trend (i.e., increasing noise level for increasing speed and thrust). The effect of the Mach number variation was evident in all cases and resulted in a difference ranging from 2.5 to

5.3 dB, as the landing configuration was progressively set. The dependence on thrust was stronger for configuration Clean and decreased for the rest of the configurations, as the contribution of the airframe noise became more dominant.

Configuration 2, with the landing gear retracted, suggests a relatively high dependence on both the Mach number and thrust and is presented in more detail by looking at the effect on the individual noise sources, as shown in Fig. 16. It is apparent that the airframe noise is the strongest contribution, which explains the high Mach number influence, as all airframe noise component models scale with U^6 [43,45], where U is the aircraft speed. This scaling is also supported well by experimental studies [66,69,70]. Interestingly, as shown in Figs. 10 and 12, the measurement data of this study indicate a lower dependency on the speed. This difference could perhaps be explained by differences in the definition of the landing gear geometry in the model and installation effects, which are not included in the prediction, but could result in spectral deviations, for example, due to interaction between landing gears and high-lift devices [71]. The speed variation also affects all the other components, with the exception of the combustor, as the cycle conditions at this component are almost unaffected. The highest contribution following the airframe component is the jet noise followed very closely by the fan discharge and turbine components. The jet noise also shows the highest variation compared to the rest of the engine components, which is attributed to the mass flow and jet velocity variation.

As expected, the change in thrust did not have any effect on the airframe noise. The jet noise increases with increasing thrust, reflecting an increase in the mass flow and jet velocity. The thrust variation directly influences the fan rotational speed, which drives the fan noise trend, together with the mass flow rate, whereas the variation in turbine noise is driven by the core mass flow and relative tip velocity

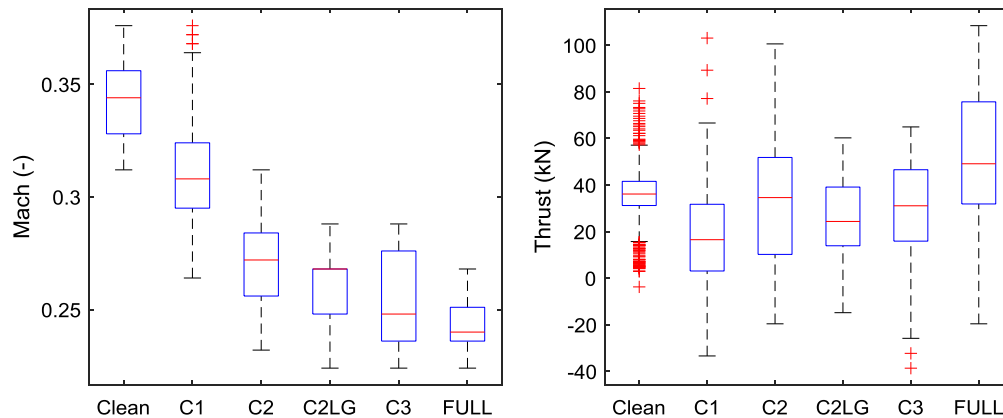


Fig. 14 Variation of the aircraft Mach number and thrust for each configuration.

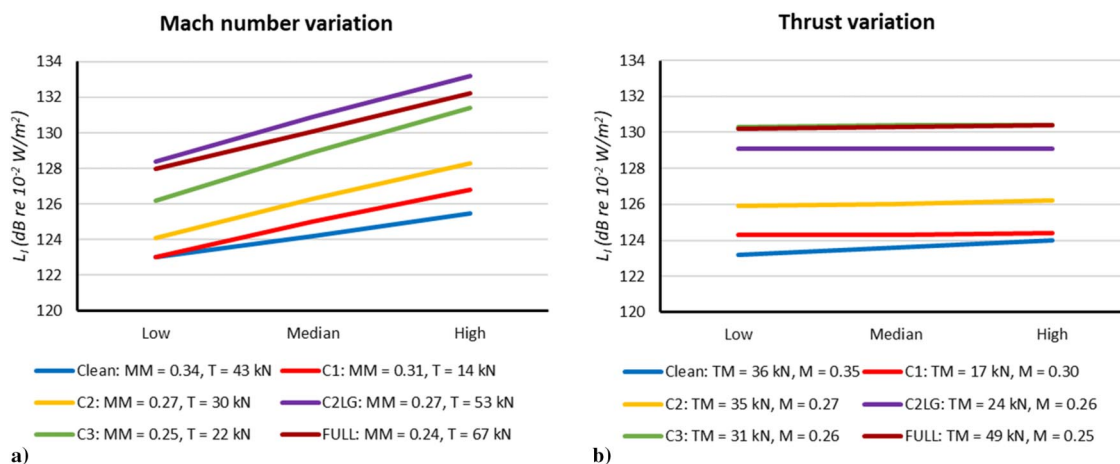


Fig. 15 Variation of total sound intensity level for 10% variation of Mach and thrust, relative to the baseline cases: a) the median Mach-number value is indicated by MM, and T is the thrust used for all cases (low, median, and high) in one configuration; the median thrust in b) is symbolized by TM and the Mach number for all cases in one configuration by M.

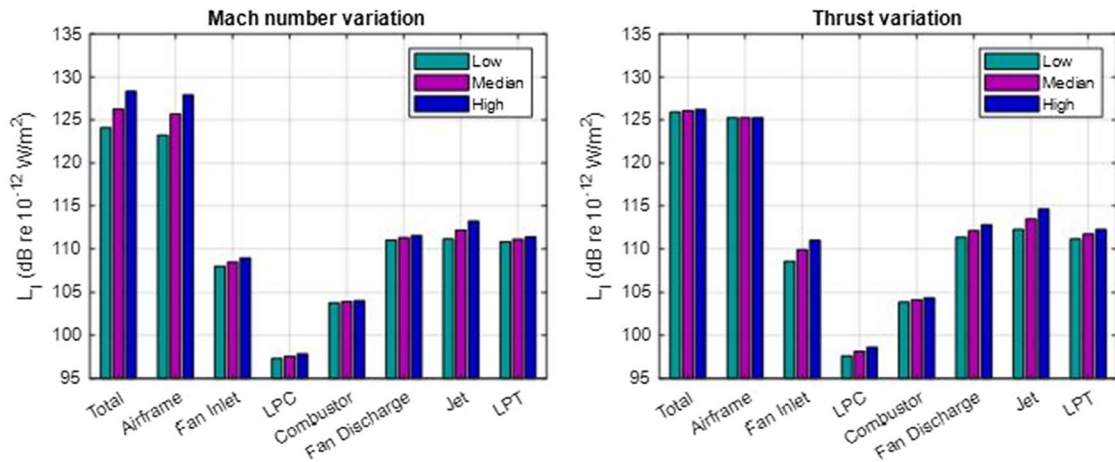


Fig. 16 Component noise level variation with flight Mach number and engine thrust for configuration 2 with the landing gear retracted; LPC = low-pressure compressor.

of the last stage. As in the previous case, the effect of thrust variation on combustor noise is negligible. A similar performance was observed for all configurations; hence, the detailed figures were not included in the analysis.

VII. Conclusions

A validation case for the source noise prediction using empirical models has been presented using FDR data and noise measurements for the A321neo. The model prediction generally showed good agreement with the measurements, especially with regard to the noise level and shape of the spectra in the mid-frequency region. It was observed that the predicted noise level was strongly dependent on the aircraft speed for all configurations. In the measurements, this effect was also evident but not to the same degree. The lack of geometric fidelity of the airframe components, especially the details of noise reduction measures, and not accounting for installation effects could partly explain this difference. When comparing the predicted and measured levels, configuration 2, without the landing gear, and configuration FULL are in good agreement. For configuration 2, with the landing gear extended, and configuration 3, the prediction and measured data matched well for frequencies above 200 Hz. Configurations Clean and 1 are relatively well predicted, especially in terms of the total noise level, while some deviations occur in the spectral shapes. Finally, the predicted noise level on the ground indicated overall good agreement with the measurements. In most cases, the predicted forward radiated noise showed greater deviation from the measurements compared to the aft radiated noise, as was also expected from previous studies [68].

One of the main reasons for the occurrence of deviations is believed to be that most of the implemented noise models are quite old, and their development was based on data from older aircraft and engines. As technology advances, noise suppression mechanisms are used, which are difficult to model. Some studies have been published over the years on updating older models to account for acoustic treatment and noise suppression, but these did not consider all noise components and were also developed before the 2000s. It is important that the models are updated continuously to match the current technology. Another source of deviation that mostly concerns the airframe model is that geometrical details of the different components are required. These are usually not publicly available, especially for the landing gear component, and approximations need to be made, resulting in possible errors in the prediction. Despite these drawbacks of the models, the noise level was well predicted for the current state-of-the-art aircraft with a high-bypass turbofan engine. At this point, it is important to note that the presented estimation concerns only the mentioned aircraft. Adjustments must be made for other aircraft and engine configurations. The directivity of various noise sources also plays an important role in aircraft noise propagation. While the presented models are able to provide directivity details [38–44], for future aircraft concepts, the

importance of directivity must be realized, and corrections should be considered for, for example, different engine installations.

The noise measurements were collected during one morning from two aircraft performing approach flybys. This created some limitations on the available data. For example, during this limited time span, the variation in atmospheric conditions was relatively small. Further studies of flights would be required to examine if these conditions have a significant effect on the measured and predicted noise, although other studies have shown that it is not expected that there will be any significant variation caused by atmospheric conditions. Furthermore, as mentioned earlier, the measurement data for some configurations are believed to be affected by errors and inaccuracies, mainly due to the distance of the aircraft from the microphone. Finally, as the data concern only approach procedures where the engine operates at a lower speed, the airframe noise became dominant, as was also observed in [35]. This means that further validation would be required for the engine noise components under conditions where they are the dominant sources (e.g., during takeoff).

The sensitivity analysis showed that the simulation model could predict variations in the noise level, even for slight variations in the operating conditions. As expected, the flight speed was more relevant for the approach segment studied here, whereas variation in the thrust setting had only a minimal effect on noise. This sensitivity to changes in operating conditions indicates that the tool can be used to assess the impact of noise during the development of new procedures.

Despite the limitations and drawbacks, it was shown that empirical and semi-empirical models can be used to predict the noise from a state-of-the-art aircraft, namely, A321neo, with good accuracy. CHOICE has been under development in Chalmers University of Technology for several years, and it has been validated for the predicted noise at the certification points on the ground. With the validation presented in this study, it is believed that CHOICE is a strong tool that can provide an accurate noise estimation in terms of spectral shape and overall sound intensity level for varying approach conditions. The comparison showed that the total sound intensity level could be predicted within an accuracy of 3 dB for most cases. The benefit of a tool, such as this, is that it has the flexibility to be applied to different scenarios from technology evaluation, as for example, in [19–21], to trajectory design [22–24]. Modifications would be required to adapt to different aircraft and engines, but that does not limit the flexibility of the models and tool.

Appendix: Source Noise Spectra and Component Breakdown

In this section, the noise source breakdown for each available configuration is presented. The spectra are presented for one point from the available trajectories, which is selected so that the aircraft speed is close to the average speed indicated in Figs. 9 and 11, for each configuration. In Figs. A1–A6, the source noise breakdown for

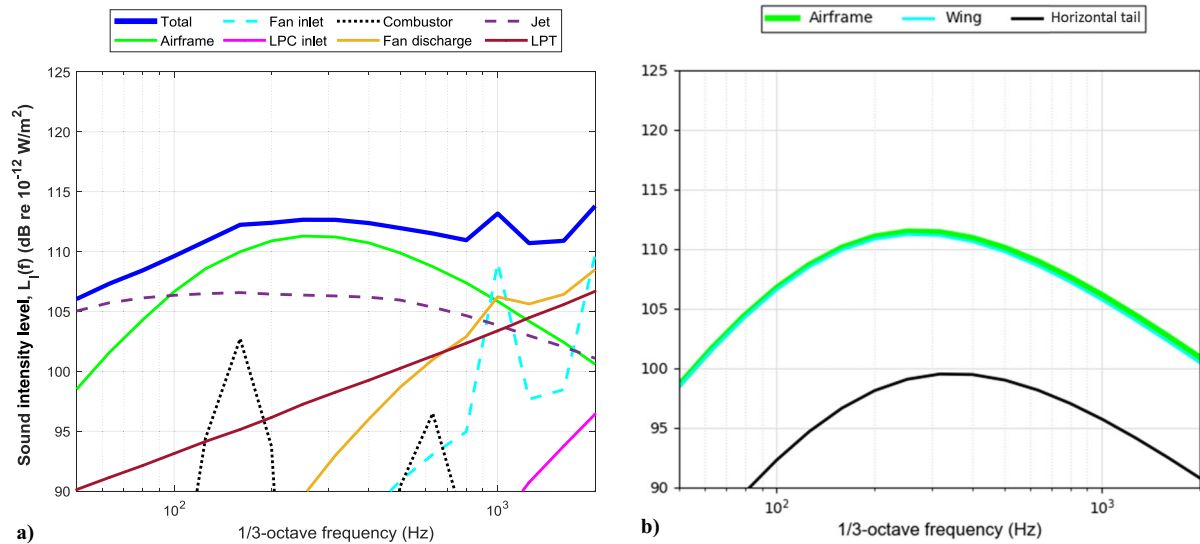


Fig. A1 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for the Clean configuration at $\theta = 90^\circ$ deg; aircraft speed at the selected point is 217 kt; LPC = low-pressure compressor.

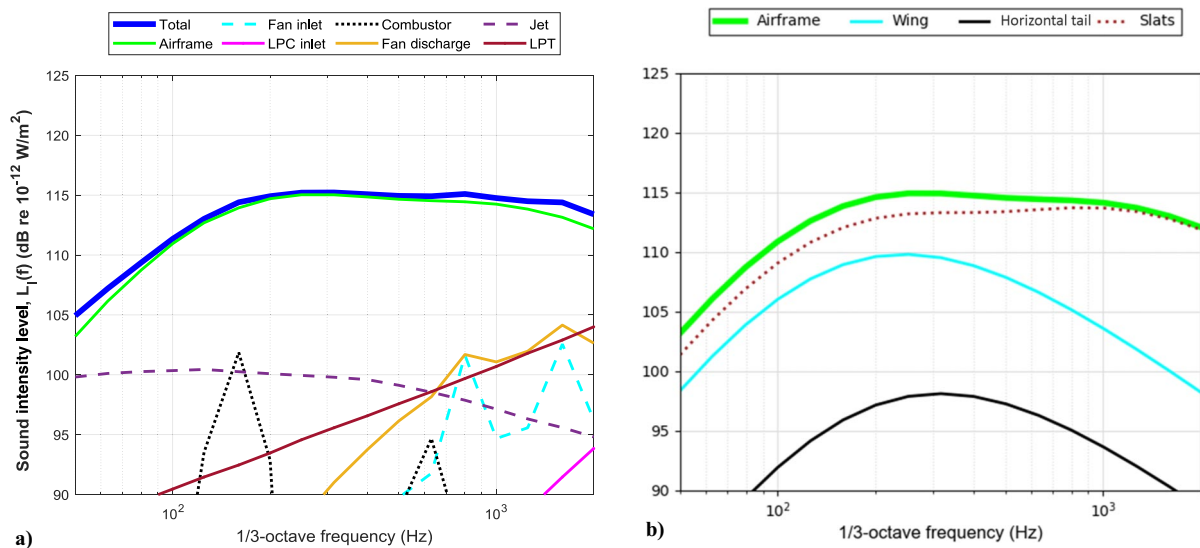


Fig. A2 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for configuration 1 at $\theta = 90^\circ$ deg; aircraft speed at the selected point is 201 kt; LPC = low-pressure compressor.

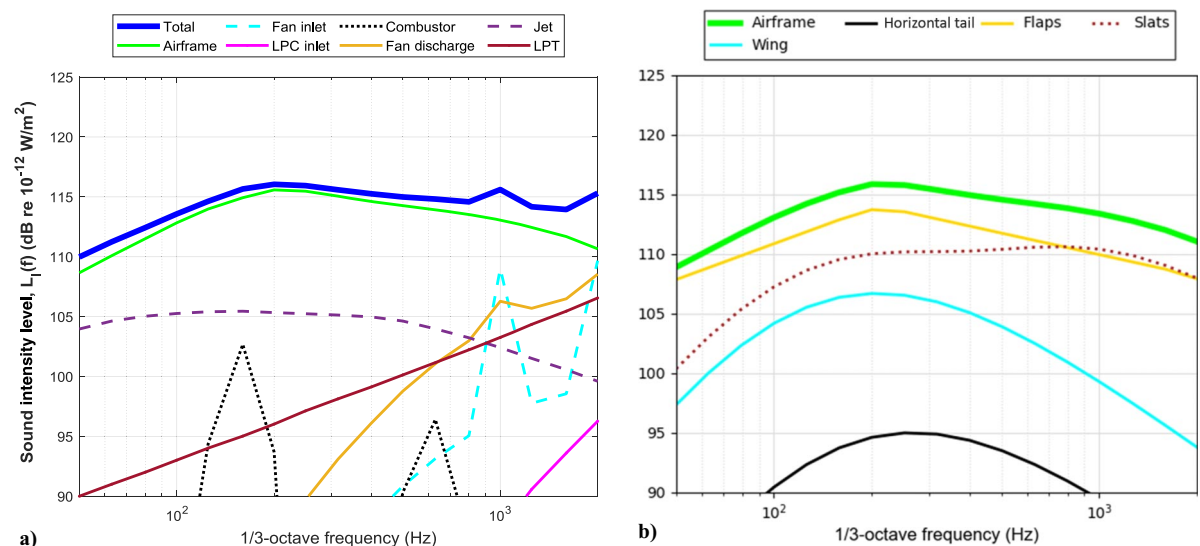


Fig. A3 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for configuration 2 at $\theta = 90^\circ$ deg; aircraft speed at the selected point is 176 kt; LPC = low-pressure compressor.

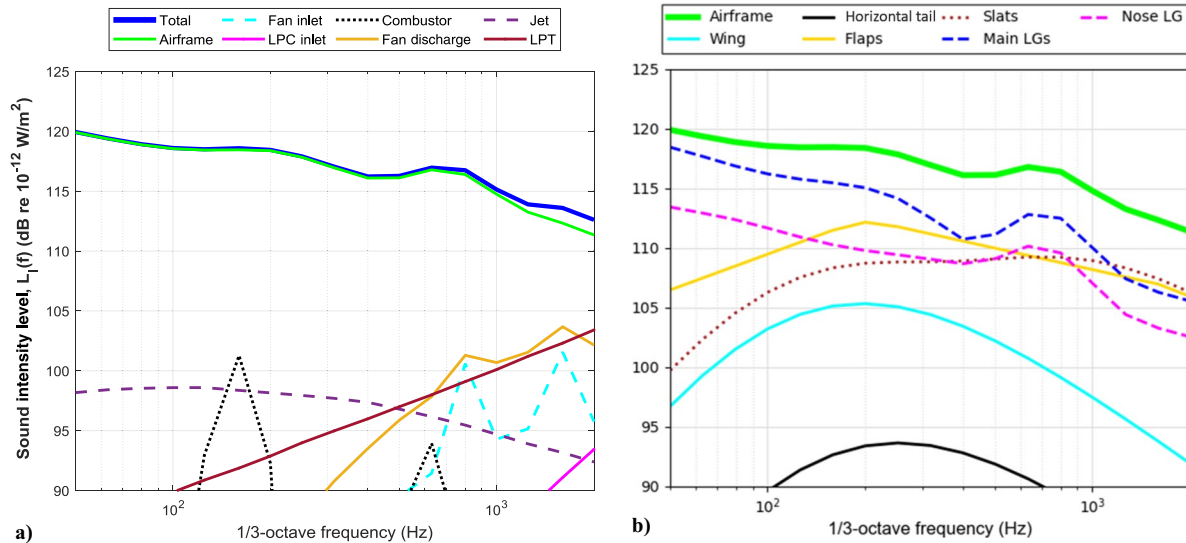


Fig. A4 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for configuration 2 with the landing gear extended at $\theta = 90^\circ$ deg; aircraft speed at the selected point is 169 kt; LPC = low-pressure compressor.

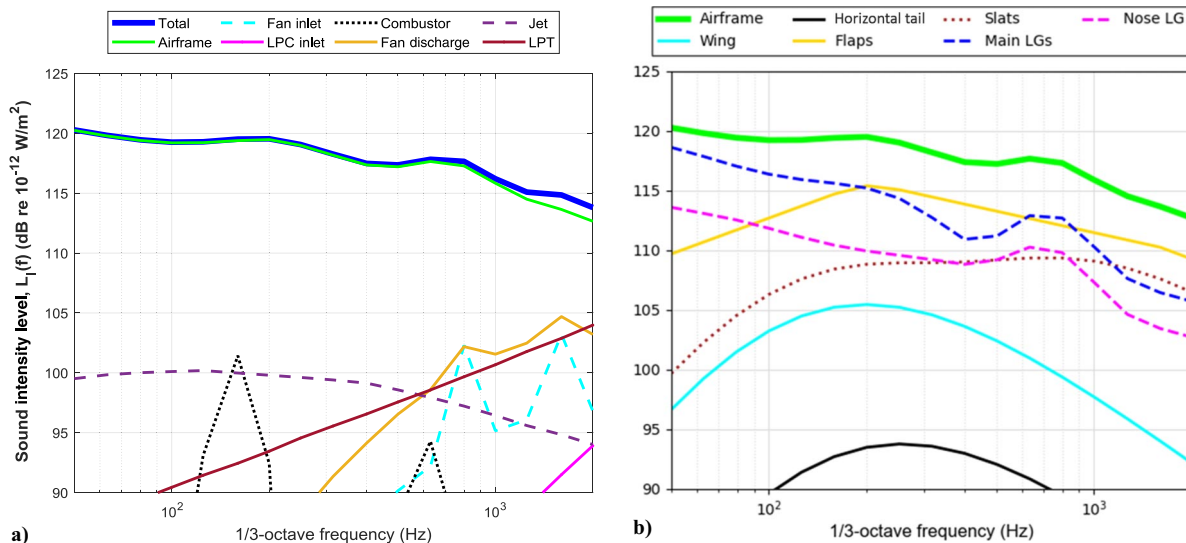


Fig. A5 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for configuration 3 (with the landing gear extended) at $\theta = 90^\circ$ deg; aircraft speed at the selected point is 168 kt; LPC = low-pressure compressor.

the entire aircraft and airframe is presented. The former includes the calculated noise level for the fan inlet, compressor, fan discharge, combustor, jet, turbine, and total aircraft, whereas the latter indicates the airframe noise breakdown, including the wing, horizontal tail, flaps, slats, main landing gears, and nose landing gear.

For all configurations, apart from the Clean configuration, the airframe noise is the dominant source, especially in low and mid-frequency regions, followed by the fan inlet, fan discharge, and low-pressure turbine (LPT) in the higher-frequency region. In the Clean configuration, jet noise becomes dominant in the low-frequency region due to the higher speed, whereas the high-frequency region is dominated by fan and LPT noise due to the increased fan rotational speed and core mass flow. The observed high LPT noise level is in line with the statement made in the Introduction that for the current-technology high-bypass ratio engines, LPT becomes an important contributor to the total engine noise [34]. However, it has also been shown that the ANOPP noise model, which was implemented here, can result in an overprediction compared to turbine noise data derived

from static engine tests [39]. For the approach conditions considered in this study, it is not possible to quantify the amount of overprediction from the model, and further comparison with measurement data where the engine noise becomes dominant would be required. Observing Figs. A1–A3, the airframe noise, as expected, gradually increases with an increase in the flap and slat angles, despite the reduction in aircraft speed. The extension of the landing gear in Fig. A4 is evident from the increase mainly in the low-frequency region. As the flaps and slats are further extended with the landing gear deployed, the airframe noise level evidently increases for the same airspeed (Figs. A4 and A5), whereas there is a slight decrease from configuration 3 to FULL (Fig. A6) due to the speed reduction. During configuration FULL, engine noise is increased as the engine is forced to spool up to counteract the drag generated by the landing gear and high flap deflection angle. The differences in the engine noise components between the configurations are attributed to the different engine operating conditions. For example, an increase in fan noise could be driven by increased mass flow or rotational speed and

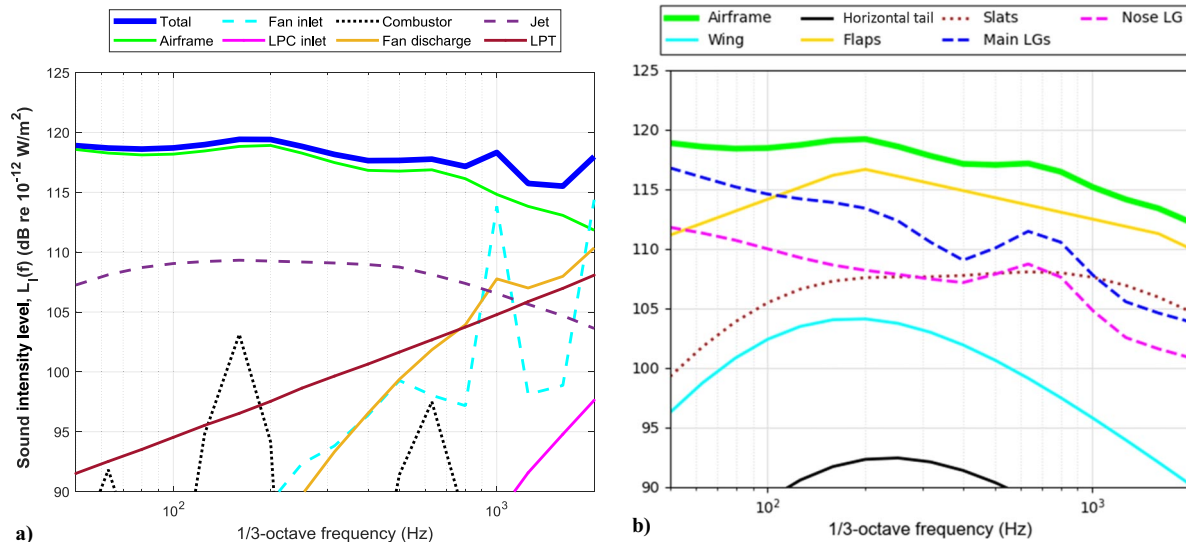


Fig. A6 Aircraft noise source spectra and component contributions for all approach configurations. a) Aircraft (engine and airframe) and b) airframe source spectrum breakdown for configuration FULL (with the landing gear extended) at $\theta = 90$ deg; aircraft speed at the selected point is 158 kt; LPC = low-pressure compressor.

in turbine noise by higher relative tip speed of the turbine's last rotor or increased core mass flow. More details on the engine performance are required to explain these differences, but this is out of the scope of this study.

Acknowledgments

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